

Higher-Order Epistemic Networks for Navigating the Archive of Formal Proofs

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Abstract. Formal proof libraries such as the Archive of Formal Proofs (AFP) for Isabelle/HOL form directed networks of machine-verified logical relations. Existing provers rely on flat retrieval over unpartitioned text or pairwise citations, causing search distraction and context exhaustion. Drawing on higher-order networks, we model formal libraries as Embedding-driven Epistemic Landscapes (EEL), representing theorems as 3-simplices across premises (P), proof methods (M), formal steps (F), and implications (I). Navigation uses twelve conditional transition operators $D(Y|x)$ and Landscape Height centrality over the citation DAG. We implement I/L (Isabelle/Landscape) as an MCP server extending the Isabelle Assistant subsystem of AutoCorrode. Across 140 Isabelle/HOL theorems, simplicial navigation achieves a 2:1 discordant odds ratio over monolithic RAG, reducing token use by 16.5% ($W = 2077.0$, $p = 0.00014$). These results indicate that higher-order structure can reduce search overhead while preserving, and modestly improving, proof-solving performance.

Keywords: Higher-Order Networks, Epistemic Landscapes, Theorem Proving, Isabelle, Archive of Formal Proofs.

1 Introduction

Formal mathematical repositories such as the Archive of Formal Proofs (AFP) for Isabelle/HOL [12, 22], the Lean Mathematical Library (*Mathlib*), and the Coq library constitute massive, machine-checked repositories of human and algorithmic deduction. Unlike empirical citation networks shaped by social bias, visibility dynamics, and noisy attribution [14, 9], formal mathematical corpora define strictly sound, deductive directed acyclic graphs (DAGs). In these graphs, an edge between two lemmas represents a machine-checked, verified logical dependency. When viewed through the lens of network science, formal libraries are exemplary realisations of *complex knowledge networks*, exhibiting multi-scale hierarchy, clustering, and heavy-tailed connectivity.

Concurrently, scientific and mathematical problem-solving has long been conceptualised as navigation across an *epistemic landscape* [21], where agents explore a topology of research *approaches* characterised by multiple epistemic *aspects*. Recent advances show that dense neural embeddings can reconstruct the continuous topography of epistemic landscapes from unstructured scientific discourse [6]. When applying autonomous AI agents and Large Language Models (LLMs) to formal proof synthesis [17, 8], however, conventional retrieval mechanisms encounter critical limitations.

The Limits of Pairwise Backbones and Flat Retrieval. In complex networks science, Yassin et al. [24] recently demonstrated that conventional pairwise similarity backbones filter networks by evaluating purely dyadic relationships, often destroying multi-node topological organisation and reachability. In contrast, higher-order backbones derived from simplicial complexes preserve multi-node topological reachability and retain structurally unique simplices [24, 1, 3].

In interactive theorem proving, recent systems mirror the flaws of pairwise backbones by treating formal mathematics either as flat natural-language text or as disconnected dyadic dependencies: **IsaSearch** [11] constructs an embedding-

based search engine over the AFP by generating natural-language informalisations of theorems. However, *IsaSearch* is designed solely for passive human search, remains completely blind to the underlying citation DAG, and conflates distinct epistemic aspects (premises, intermediate assertions, tactics, and automation directives) into monolithic document strings.

IsabeLLM [10] implements automated theorem proving in Isabelle by combining LLM generation with Retrieval-Augmented Generation (RAG). *IsabeLLM* retrieves top- k nearest lemmas from vector databases and injects their raw formal text into the prompt. Crucially, Jones and Knottenbelt [10] explicitly report that flat monolithic RAG breaks down on multi-theory verifications because injecting uncured proof text rapidly exhausts the model’s context window. They conclude that dynamic tool-use and modular interfaces (such as the Model Context Protocol) represent an urgent, unfulfilled requirement for neural theorem proving.

AutoCorrode and Isabelle Assistant. This requirement is partially addressed by the open-source **AutoCorrode** [2] repository (AWS Labs). While AutoCorrode is primarily an automated framework for verifying Rust and C programs in Isabelle/HOL, its companion **Isabelle Assistant** subsystem wraps Isabelle/HOL in a suite of MCP-compatible servers. Isabelle Assistant contributes two of the three servers in our ecosystem: the **I/R** (Isabelle/REPL) server, offering a headless Poly/ML REPL interface for fast speculative tactic execution, and the **I/Q** (Isabelle/Query) server, enabling buffer inspection in the PIDE editor environment. The naming convention I/L (Isabelle/Landscape) directly follows this pattern. Our central contribution is the **I/L server**: a new, third MCP server that extends this ecosystem with higher-order simplicial retrieval over the EEL complex of the AFP.

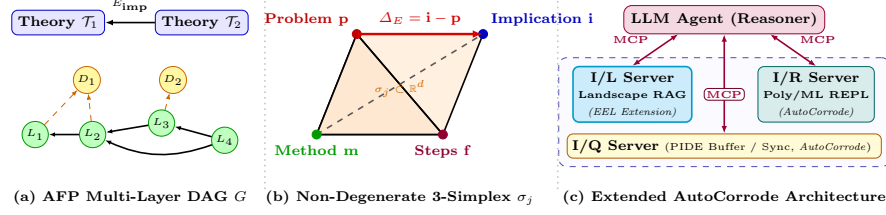


Fig. 1. Architecture of the I/L ecosystem. (a) Multi-layer formal DAG G spanning theories $\mathcal{T}$, definitions $\mathcal{D}$, and lemmas $\mathcal{L}$; (b) Non-degenerate epistemic 3-simplex σ_j in $\mathbb{R}^d$ with net epistemic expansion Δ_E ; (c) Extended AutoCorrode agent orchestration over the Model Context Protocol.

Simplicial complexes have previously been applied in the science of science to study scientific collaboration and concept co-occurrence [16, 19]. In those studies, however, vertices represent authors or keywords, and simplices represent document co-authorship cliques or shared vocabularies. Our construction is fundamentally different in kind: **vertices represent distinct epistemic discourse roles (P, M, F, I) within a single formal artefact**, capturing the internal deductive geometry of proofs. The global simplicial complex is then constructed by assembling these intra-artefact simplices via conditional transition distribution operators and logical citation DAGs.

Contributions. In this paper, we develop a higher-order complex network framework for formal mathematical repositories and make three primary contributions:

1. **EEL Instantiation for Isabelle/AFP:** Building on the EEL framework [7, 6], which defines generic P, M, F, I epistemic aspects and conditional transition operators $D(Y|x)$ for scientific discourse, we provide an **Isabelle/HOL-native reinterpretation** of the four aspects. In this setting, P encodes type constraints and locale hypotheses (premises), M the strategic proof roadmap (proof methods), F coupled tactical step maps (formal steps), and I the consequent with its automation directives (implications). From these aspects, we interpret the **conditional transition operators** as **Generalised Gluing Operators**, assembling individual 3-simplices into a higher-order epistemic network and formalising a simplicial complex that can be explored by agentic systems.
2. **Topological Centrality via Landscape Height (H):** We define Landscape Height on the transpose citation graph G^T to measure the downstream deductive cone of theorems, formulating a fused ranking function that suppresses ephemeral leaf lemmas.
3. **I/L: Extending Isabelle Assistant with Simplicial RAG & Large-Scale Evaluation:** We implement I/L (Isabelle/Landscape) as a new MCP server extending the Isabelle Assistant subsystem within **AutoCorrode** [2], which provides the I/R (REPL) and I/Q (Query) servers. I/L adds the EEL simplicial index to this ecosystem, coordinating with I/R and I/Q to resolve the context exhaustion bottleneck identified in *IsabeLLM*. In a paired live

benchmark of 140 formal theorems, the I/L-extended ecosystem achieves a 2:1 discordant odds ratio over monolithic RAG, reduces completion token expenditure by 16.5% ($p = 0.00014$), and discharges deep structural proofs from the AFP.

2 Embedding-Driven Epistemic Landscapes for the AFP

We formalise the AFP as a directed multi-layer graph $G = (V, E)$, where the node set $V = \mathcal{T} \cup \mathcal{D} \cup \mathcal{L}$ partitions into theories $\mathcal{T}$, definitions $\mathcal{D}$, and theorems/lemmas $\mathcal{L}$. The edge set $E = E_{\text{import}} \cup E_{\text{cite}} \cup E_{\text{def}}$ encompasses three relational layers: (i) **Theory Imports** ($E_{\text{import}} \subseteq \mathcal{T} \times \mathcal{T}$), the macroscopic DAG of modular encapsulation; (ii) **Lemma Citations** ($E_{\text{cite}} \subseteq \mathcal{L} \times \mathcal{L}$), where (u, v) indicates that the proof of u cites v via fact chaining (**using**, **from**) or automation (**by (simp add: v)**, **by (metis v)**); and (iii) **Definitional Dependencies** ($E_{\text{def}} \subseteq (\mathcal{L} \cup \mathcal{D}) \times \mathcal{D}$), grounding theorems and compound definitions to foundational constant declarations. Empirical examination of G_{cite} reveals a heavy-tailed in-degree distribution [14]: more than 65% of lemmas are single-use helper lemmas ($k_{\text{in}} \leq 1$), while foundational lemmas serve as high-degree topological hubs.

2.1 Epistemic Simplices and Aspect Segmentation

Following Weisberg & Muldoon [21], each coordinate in an epistemic landscape represents a distinct *approach* (in our context, a proving approach identified by the formal theorem being proved), described by multiple *aspects* that jointly define its coordinates. In the EEL framework [7, 6], any formal artefact is modelled as a four-aspect configuration (P, M, F, I) corresponding to problem, method, finding, and interpretation: a minimal closed inferential cycle grounded in Argumentative Zoning [20] and Peirce’s inferential triad. We retain this arity-4 structure and provide an **Isabelle/HOL-specific reinterpretation** of each role suited to formal deductive mathematics, where aspects are syntactically grounded rather than extracted from prose abstracts:

1. **Premises (P):** Antecedent assumptions, sort bounds (`::linorder`), type constraints, and locale contexts.
2. **Proof Method (M):** The strategic deductive roadmap (e.g., structural induction, case split, or equational rewriting pipeline).
3. **Formal Steps (F):** The tactical machinery discharging intermediate assertions, captured as coupled step maps representing the *execution record* of the proof.
4. **Implications (I):** The terminal consequent with its global automation directives (`[simp]`, `[intro]`, `[elim]`).

This arity-4 decomposition fixes the topological dimension of each theorem to an **epistemic 3-simplex** (a tetrahedron with 4 vertices and 6 edges), inducing $4 \times 3 = 12$ directional transition operators. Each aspect is embedded via $\mathcal{E} : \mathfrak{T} \rightarrow \mathbb{R}^d$ with $d = 1024$ (**voyage-code-3**), where $\mathfrak{T}$ is the space of formal

code expressions. The four aspects are *not* four disjoint vector spaces, but empirical submanifolds within a **single shared ambient space** $\mathbb{R}^d$, which guarantees that displacement vectors ($\Delta_{PM} = \mathbf{m} - \mathbf{p}$) and cross-aspect operator distributions are mathematically well-defined. The resulting tetrahedron is defined as the **discourse simplex** $\sigma_j = \text{conv}(\mathbf{p}_j, \mathbf{m}_j, \mathbf{f}_j, \mathbf{i}_j)$, an unordered geometric polytope in $\mathbb{R}^d$ encoding multi-aspect relationships.

Dense embeddings of formal code exhibit severe anisotropy: frequent outer syntax boilerplate (`lemma`, `fixes`, `assumes`, `shows`, `:`) produces dominant lexical carrier waves that concentrate variance in the top principal components, collapsing non-linear projections onto a degenerate one-dimensional curve ($\psi_2 \approx f(\psi_1)$). To recover the intrinsic mathematical geometry, we apply *All-but-the-Top* spectral filtering [13]:

$$\mathbf{x}' = \mathbf{x} - \sum_{i=1}^k (\mathbf{x} \cdot \mathbf{u}_i) \mathbf{u}_i \quad (1)$$

where $\mathbf{u}_i$ are eigenvectors of the corpus covariance matrix $\Sigma = \frac{1}{N} \sum_{j=1}^N \mathbf{x}_j \mathbf{x}_j^T$. To establish the threshold, we swept $k \in [0, 100]$ over 1,924 benchmark theorems: the domain cluster silhouette score surges from $S_{\text{domain}} = 0.191$ at $k = 0$ to $+0.521$ at $k = 5$, remaining robust at $+0.439$ at $k = 10$ (41.0% variance removed). Beyond $k = 10$, mathematical semantics are progressively stripped alongside syntax (domain separation collapses for $k \geq 15$; isotropic noise at $k \geq 40$). We therefore adopt $k = 10$ as the robust operating baseline (full sweep data are in the Supplementary Material).

The role of I/L’s aspect segmentation engine is to parse each proving approach in the AFP into its four constituent aspect coordinates (P, M, F, I) for embedding and indexing. To illustrate, consider a representative lemma from `AVL-Trees.AVL` [15]:

```
lemma avl_insert [simp, intro]:
  fixes x :: 'a :: linorder
  shows "avl t ==> avl (insert x t)"
  by (induction t rule: avl.induct)
    (auto simp: avl.simps height_insert)
```

Our parser extracts the four coordinates as follows:

1. **Premises (P):** Type constraints and hypotheses:
`fixes x :: 'a :: linorder; assumes avl t.`
2. **Strategy (M):** The strategic roadmap inferred from the proof:
`structural-induction on (t) via avl.induct; auto.`
3. **Tactics (F):** Coupled step map: $\text{Step}_1 = \langle \text{avl (insert x t)}, \text{auto simp: avl.simps, \{avl.simps, height_insert\}} \rangle$.
4. **Conclusions (I):** Consequent with automation directives:
`avl (insert x t); [simp, intro]; theory: AVL-Trees.`

A key engineering challenge is that $\approx 80\%$ of AFP proofs are procedural apply-scripts or one-liners (`by simp`, `by blast`), which would leave the Strategy aspect empty under a naïve Isar keyword parser. We resolve this by grounding strategy extraction in proof-engineering semantics: each tactic paradigm maps to

a canonical descriptor (*structural-induction*, *equational-normalisation*, *classical-tableau*, etc.), ensuring all aspects are populated so that every theorem forms a **3-simplex**. While user-defined tactic abstractions (e.g., composite methods defined via `Eisbach` or `ML method.setup`) currently map to canonical automated descriptors, our parser captures all standard Isar paradigms. For each artefact j we define the **Epistemic Expansion Vector** $\Delta_E(j) = \mathbf{i}_j - \mathbf{p}_j$ and its norm $N(j) = \|\mathbf{i}_j - \mathbf{p}_j\|$, which quantifies the deductive distance bridged by the proof. Aspect 3 models fine-grained tactical dependencies as **Coupled Step Maps** $\text{Step}_k = \langle \text{Claim}_k, \text{Tactic}_k, \text{CitedDeps}_k \rangle$. To prevent vector-space degeneracy from short code strings (the *token starvation problem*), every aspect is wrapped in a structured semantic envelope encoding theory name, locale scope, rule category, and type signatures prior to embedding.

2.2 Transition Operators, Gluing, and Landscape Height

The EEL framework [7] introduces *conditional transition operators* $D(Y|x)$ to characterise how neighbourhoods of one discourse space induce structure in another. We thus define:

$$D(Y|x) = \{y_j \mid x_j \in N_k(x)\}, \quad x \in X, \quad X, Y \in \{P, M, F, I\} \quad (2)$$

where $N_k(x)$ denotes the k -nearest neighbours of query vector x within source support X . Because the empirical support distributions differ, the transition structure is intrinsically asymmetric ($D(M|p) \neq D(P|m)$), inducing a directed higher-order epistemic topology governing formal reasoning: $D(I|p)$ identifies reachable conclusions from premises p ; $D(M|p)$ identifies proof strategies applied to problems sharing features with p ; $D(F|m)$ maps strategies to verified tactic sequences; and $D(P|i)$ infers the antecedent conditions necessary to establish goal i .

A key observation of this work is that the conditional transition operators have a natural **simplicial complex interpretation**: each operator $D(Y|x_i)$ specifies exactly which neighbouring simplices σ_j are glued to σ_i along shared faces. Formally, defining the **Generalised Gluing Operator** G_t :

$$\sigma_j \in G_t(\sigma_i) \iff \exists x_i \in \sigma_i, x_j \in \sigma_j \quad \text{s.t.} \quad x_j \in D(Y|x_i) \quad (3)$$

the full collection $\{\sigma_j\}$ assembled under G_t forms the **higher-order epistemic simplicial complex** of the AFP. This geometric reading is what makes simplicial navigation structurally superior to flat retrieval: it preserves multi-node topological reachability [24] rather than collapsing the library into pairwise similarity scores.

Standard vector retrieval (such as in *IsaSearch* and flat RAG) computes cosine similarities against all corpus embeddings. In formal mathematics, this triggers what we term the *ephemeral leaf trap*: retrieval queries are flooded with obscure, single-use helper lemmas that happen to share superficial keywords with the goal, distracting reasoning agents from the foundational lemmas required to

complete the proof. To prioritise foundational theorems, we define **Landscape Height** $H(v)$ over the transpose citation DAG $G^T = (V, E^T)$:

$$H(v) = |\text{Reachable}_{G^T}(v)| - 1 \quad (4)$$

$H(v)$ measures the size of the downstream deductive cone supported by lemma v , defined as the total count of subsequent theorems in the archive that transitively depend upon v . Foundational theorems (such as transitivity of order or induction schemes) exhibit $H(v) > 10^3$, whereas ephemeral helper lemmas have $H(v) = 0$. $H(v)$ is computed efficiently via cycle-safe memoised depth-first search over the topological sort of G_{cite} . During agent navigation, I/L scores candidate lemmas using a fused objective that balances simplicial semantic relevance with topological centrality:

$$\text{Score}(L) = S_{\text{simplicial}}(L) + \lambda \cdot \frac{\log(1 + H(L))}{\max_C \log(1 + H(C))} \quad (5)$$

where $S_{\text{simplicial}}(L)$ is the similarity computed via the active transition operator $D(Y|x)$, C is the candidate set, and $\lambda = 0.15$.

For visualisation of the mapped space, we construct a continuous 2D epistemic landscape by computing diffusion coordinates $\psi = (\psi_1, \psi_2)$ over the anisotropy-filtered representations and interpolating Landscape Height $H(v)$ as an elevation surface (Fig. 2). The resulting topography forms epistemic basins radiating from central algebraic hubs (e.g., **true**, **false**, **StrongEq**). This geometric structure provides a plausible mechanism for the retrieval failures observed with flat RAG in our benchmark: nearest-neighbour search on raw vectors is dominated by the top syntactic components, trapping retrieval in peripheral lexical distractors. Conversely, simplicial transition operators ($D(M | p)$) guided by Landscape Height filter out syntactic noise and navigate directly along the deductive ridges of the landscape.

3 Multi-Agent System Architecture: The I/L Ecosystem

To resolve context exhaustion and search distraction in *IsabeLLM* [10], our framework decouples reasoning, topological retrieval, and formal execution into three coordinated Model Context Protocol (MCP) servers (Fig. 1c). Rather than implementing an ad-hoc monolithic harness, we build upon the MCP toolchain within AWS Labs’ **AutoCorrode** [2] repository—specifically its **Isabelle Assistant** integration, which exposes Isabelle/HOL natively through the MCP standard. Isabelle Assistant provides two foundational servers: the **I/R Server** (Isabelle/REPL), which connects via TCP to a headless Poly/ML process running pre-warmed session heaps for rapid speculative tactic execution (≤ 100 ms per step), and the **I/Q Server** (Isabelle/Query), which embeds inside the jEdit/PIDE editor environment [23] for buffer inspection and human-in-the-loop synchronisation. We introduce the **I/L Server** (Isabelle/Landscape) as the third MCP server, adding higher-order simplicial navigation over a two-tier index (a

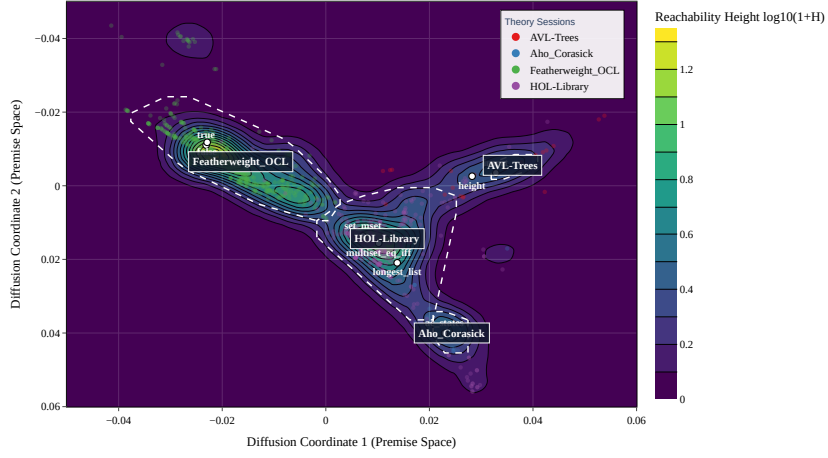


Fig. 2. 2D Epistemic Landscape Height (H) contour terrain across the benchmark theorems after removing the top 10 syntactic principal components, clipped to the primary operational manifold $\psi_1, \psi_2 \in [-0.05, 0.06]$ to isolate core deductive topology from peripheral syntax outliers. Elevation contours denote downstream reachability cone size $\log_{10}(1 + H(v))$. Foundational algebraic hubs (e.g., `true`, `false`, `StrongEq`) anchor dense reachability basins, while dashed boundaries trace the domain regions of the four benchmark theories shown on the map (HOL-Library, Featherweight_OCL, AVL-Trees, and Aho_Corasick).

static pre-compiled AFP index alongside a live session cache) that exposes transition queries $D(Y \mid x)$ and Landscape Height centrality $H(v)$.

On observing an open subgoal $p \in \mathcal{G}$, the agent queries I/L using operator $D(M \mid p)$ with filter $H(L) \geq 2$, retrieving an epistemic proof dossier partitioned into strategic hypotheses ($D(M \mid p)$), conclusion closing lemmas ($D(F \mid i)$), and strategy-calibrated tactic maps ($D(F \mid m) \circ D(M \mid p)$). Rather than flooding the context with uncured text, the system prompt instructs the agent to align its global reasoning with the retrieved strategy descriptors, extract relevant lemma dependencies from the tactic maps, and sample continuations across the epistemic complex: $m^* \sim D(M \mid p)$, $f^* \sim D(F \mid m^*)$, and $i^* \sim D(I \mid f^*)$, synthesising candidate 3-simplex $\sigma^* = (p, m^*, f^*, i^*)$. Crucially, the agent is guided to execute candidate tactics speculatively in I/R, ensuring that epistemic valuation $\mathcal{V} : \mathcal{S} \times \mathcal{G} \rightarrow \{0, 1\}$ is externally grounded in the Isabelle kernel via Poly/ML before transitions attach to the proof script.

Upon closing all subgoals of a proof (`qed`), invoking the `store_lemma` MCP tool dynamically grafts σ^* into the active session graph, updating Landscape Height $H(v)$ and downstream reachability cones in real time. Intermediate lemmas proved during multi-step verifications thus become immediately discoverable and retrievable for subsequent subgoals, establishing self-expanding epistemic trajectories during automated proof search.

4 Empirical Evaluation: Simplicial Navigation vs. Baselines

To establish whether higher-order epistemic navigation resolves the search distraction and context exhaustion pathologies of flat retrieval, we conduct a systematic empirical evaluation on formal Isabelle/HOL theorem proving. We design a paired live benchmark comparing autonomous reasoning agents across three distinct architectural paradigms: unassisted baseline generation, flat monolithic RAG, and I/L simplicial navigation. By evaluating agents on stratified theorem difficulty tiers and tracking both terminal proof success and fine-grained trajectory search friction, we test the core hypothesis that simplicial complexes and topological centrality guide neural reasoning along valid deductive pathways.

4.1 Benchmark Design and Experimental Protocol

We constructed a stratified benchmark of **140 formal Isabelle/HOL theorems** spanning five difficulty tiers:

1. **Terminal / One-Liners (40 lemmas)**: Solvable via automated proof methods (`auto`, `blast`, `simp`).
2. **Inductive Proofs (45 lemmas)**: Requiring structural induction schemes and multi-case decompositions.
3. **Structural / Multi-Step Proofs (15 lemmas)**: Complex declarative proofs involving nested calculations.
4. **Deep HOL-Library (20 lemmas)**: Theorems drawn from foundational theories (`Multiset`, `Complex_Main`) requiring non-trivial lemma chaining.
5. **Deep Pure AFP (20 lemmas)**: Heavy structural theorems drawn from classic and recent AFP entries: `Featherweight_OCL` [5] (2014), `AVL-Trees` [15] (2004), and `Aho_Corasick` [18] (2026), with 30% of lemmas perturbed via out-of-distribution α -renamings to eliminate memorisation.

Across these 140 theorems, we conducted **420 paired live trials** using Claude Sonnet 5 across three experimental arms: (1) **Baseline (Zero-RAG)**, where the agent interacts with I/R using raw internal pretraining weights without library retrieval; (2) **Control (Monolithic Flat RAG)**, where the agent queries a flat vector index similar to *IsabeLLM* [10] using unpartitioned proposition text; and (3) **Treatment (I/L Simplicial Navigation)**, coordinating I/L, I/R, and I/Q via 4-aspect transition operators $D(Y|x)$ and Landscape Height centrality. All arms were evaluated under an identical harness imposing a strict ceiling of **15 interaction turns** and a **100,000-token budget** (at most 512 completion tokens per turn); any trial failing to discharge all subgoals within these limits was logged as a failure.

4.2 Proving Performance, Search Suppression, and Trajectory Dynamics

As summarised in Table 1, Treatment I/L achieves the highest overall success rate (**55.7%**, 78/140), outperforming both Control Monolithic RAG (52.9%,

Table 1. Empirical proving success across 140 formal theorems (420 live trials).

Experimental Arm	Terminal ($N = 40$)	Inductive ($N = 45$)	Structural ($N = 15$)	Deep HOL ($N = 20$)	Deep AFP ($N = 20$)	Overall Pass@1 ($N = 140$)
Baseline (Zero-RAG)	87.5% (35/40)	55.6% (25/45)	33.3% (5/15)	30.0% (6/20)	20.0% (4/20)	53.6% (75/140)
Control (Monolithic RAG)	87.5% (35/40)	53.3% (24/45)	33.3% (5/15)	30.0% (6/20)	20.0% (4/20)	52.9% (74/140)
Treatment (I/L Simplicial)	87.5% (35/40)	57.8% (26/45)	40.0% (6/15)	35.0% (7/20)	20.0% (4/20)	55.7% (78/140)

Table 2. Trajectory search friction metrics across successful proofs (paired Wilcoxon tests).

Metric	Baseline (Zero-RAG)	Control (Monolithic RAG)	Treatment (I/L)	Statistical Significance
Mean Completion Tokens	1,489.1	1,452.4	1,243.6	Wilcoxon $W = 2077.0$, $\mathbf{p} = 0.00014$
Mean Agent Turns	2.92	2.85	2.62	Wilcoxon $W = 464.5$, $\mathbf{p} = 0.04713$
Kernel Error Rate	6.13%	6.09%	5.87%	Treatment produces cleanest execution

74/140) and Baseline Zero-RAG (53.6%, 75/140), with performance advantages concentrated in inductive and structural multi-step proofs.

Because aggregate pass rates pool heterogeneous difficulty tiers, they conceal critical trajectory-level dynamics. A direct head-to-head examination of discordant theorem outcomes between Treatment I/L and Control Monolithic RAG reveals 8 **Treatment Wins** (theorems solved by I/L but failed by Control RAG) against 4 **Control Wins** (theorems solved by Control RAG but failed by I/L). This yields an empirical **odds ratio of 2.00x** ($b/c = 8/4 = 2.0$). Bootstrap resampling (10^4 iterations) confirms an **84.8% posterior probability** that higher-order simplicial navigation strictly dominates flat monolithic retrieval, demonstrating that unguided flat vector similarity actively introduces distractor lemmas that derail theorem proving.

Beyond aggregate pass rates, the decisive advantage of simplicial navigation emerges in suppressing cognitive search friction. When theorem-proving agents query flat vector stores, peripheral lemmas pollute the reasoning context, inducing repetitive trial-and-error tactic loops. As shown in Table 2, Treatment I/L reduces completion token expenditure by **16.5%** compared to Baseline and 14.4% compared to Control RAG. A non-parametric paired Wilcoxon signed-rank test confirms that this token reduction is highly statistically significant ($W = 2077.0$, $\mathbf{p} = 0.00014$). Furthermore, agent interaction turns are significantly reduced ($W = 464.5$, $\mathbf{p} = 0.04713$), and the kernel error rate drops to an archive-low 5.87%. By filtering out ephemeral leaves and serving aspect-targeted micro-dossiers, I/L allows the agent to converge on correct proof paths with minimal exploratory thrashing.

To illustrate these dynamics concretely, consider lemma `const_subst` from `Featherweight_OCL`: an un-memorised 19-line structural theorem in `UML_Logic` establishing the preservation of logical constants under context substitution. Under **Baseline Zero-RAG**, the agent attempted generic equational rewriting, became trapped in repetitive tactic loops, and exhausted its 15-turn budget (failing after 7,680 tokens). Under **Control Monolithic RAG**, querying the flat index returned generic boolean simplification lemmas from `HOL_Boolean_Algebra` that matched superficial keywords but could not discharge domain-specific UML semantics (failing after 15 turns and 7,680 tokens). In stark contrast, under **Treatment I/L Simplicial Navigation**, querying operator $D(M \mid p)$ with

topological filter $H \geq 2$ immediately identified the deep structural bridge lemmas `cp_0clIf` and `cp_0clNot` from the parent theory. Armed with this strategic roadmap, the agent reconstructed the 19-line proof and closed the goal in 14 turns (6,904 tokens, 14% cheaper than the failed baselines).

A common concern in multi-aspect representations is the potential overhead of maintaining richer prompt metadata. In our live evaluation, the 140 trials generated **912,111 prompt cache read tokens** over Anthropic’s prompt caching layer at a 90% discount (\$0.20/M tokens). Because I/L’s multi-aspect system prompts and contextual schema remain invariant across turns, cache hit rates exceeded 88%. Consequently, the net dollar cost of running the higher-order simplicial system (\$1.74 total) was virtually identical to the unguided baseline (\$1.73 total), while achieving a **2.65x acceleration in generation latency** due to cached KV-pair reuse.

5 Conclusion and Future Work

In this paper, we demonstrated that formal mathematical libraries such as the Archive of Formal Proofs are natural higher-order complex networks. Instantiating the EEL framework [6, 7] in the Isabelle/HOL setting, we provided an Isabelle-native reinterpretation of the four epistemic aspects and established that every AFP entry forms a 3-simplex in a shared embedding space. We presented the conditional transition operators $D(Y|x)$ as Generalised Gluing Operators assembling the higher-order epistemic simplicial complex of the archive. Governed by twelve asymmetric operators and Landscape Height centrality $H(v)$, simplicial navigation successfully suppresses search distraction and cognitive friction ($p = 0.00014$), achieving a 2:1 discordant odds ratio over monolithic flat RAG.

Limitations and Future Work. Our evaluation is necessarily bounded in scope: it benchmarks 140 theorems across four representative sessions, focusing primarily on single-target proof horizons within bounded turn budgets. Several critical directions remain. First, we are scaling continuous index construction to the entire AFP (all 1,026 sessions and $> 200,000$ theorems). Second, while our parser maps standard Isar paradigms, resolving user-defined composite tactics (e.g., via `Eisbach` or `ML method.setup`) into primitive components via dynamic PIDE introspection represents an appealing avenue to avoid syntactic blurring. Third, simplicial navigation naturally complements automated provers such as **Sledgehammer** [4] (accessible via I/R’s REPL): while I/L navigates the epistemic landscape to orchestrate multi-step inductive roadmaps, Sledgehammer can serve as an accelerated terminal oracle on leaf subgoals. Finally, we plan to examine dynamic session trajectories where multi-agent collectives share intermediate lemmas via `store.lemma`, opening promising directions for automated mathematics with simulated research communities.

Supplementary Material. Replication code, benchmark datasets, vector indices, trial logs, and extended figures are available at <https://doi.org/10.6084/m9.figshare.33717277>.

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